

Chapter 1 — Uses of datasets

Where datasets create value—and one end-to-end workflow

Learning objectives

- Map sectors to typical datasets and analytical goals
- Trace the shared lifecycle: define, assess, explore, act
- Outline the customer-churn case study steps

Applications across sectors

- Retail: purchase histories → segmentation, forecasts
- Healthcare: EHRs and wearables → risk and monitoring
- Finance: ledgers → fraud and credit scoring
- Education and government: logs and statistics →
intervention and planning

Example 1.21 — Purchase history

```
$ cd modules/chapter1/example21/ && bash run.sh
```

Case study: customer churn

- Collect and join customer behavior features
- Explore and clean before modeling
- Train a classifier; score risk of cancellation
- Act with retention offers guided by feature importance

Example 1.31 — Churn model sketch

```
$ cd modules/chapter1/example31/ && bash run.sh
```

Takeaways

- Domain vocabulary changes; the lifecycle does not
- Exploration and quality feed every serious application
- Churn shows define → prepare → model → intervene